Gamma-ray burst classification

Andrea Marcone

Introduction to Bayesian probability theory

21/04/2026

Index

- The Gamma Ray Bursts (GRBs)
- Classification on $\log(T_{90})$
- ? The parameters
- Classifying a GRB
- Probabilities of Short and Long GRB
- The intermediate class
- Classification in the $\log(T_{90}) - \log(HR)$ plane

The Gamma Ray Bursts (GRBs)

They are transient events of high energy photons and extragalactic origin, discovered in 1973

They are usually composed of a first peak (prompt emission) and then an afterglow

We will use two proprieties to classify them:

- The T_{90}: the time needed to pass from 5% to 95% of the total flux of the prompt emission

- The HR: the ratio between fluxes in different energy bands

Tipical light curve of a GRB?
Or artistic representation?

Classification on $\log(T_{90})$: data

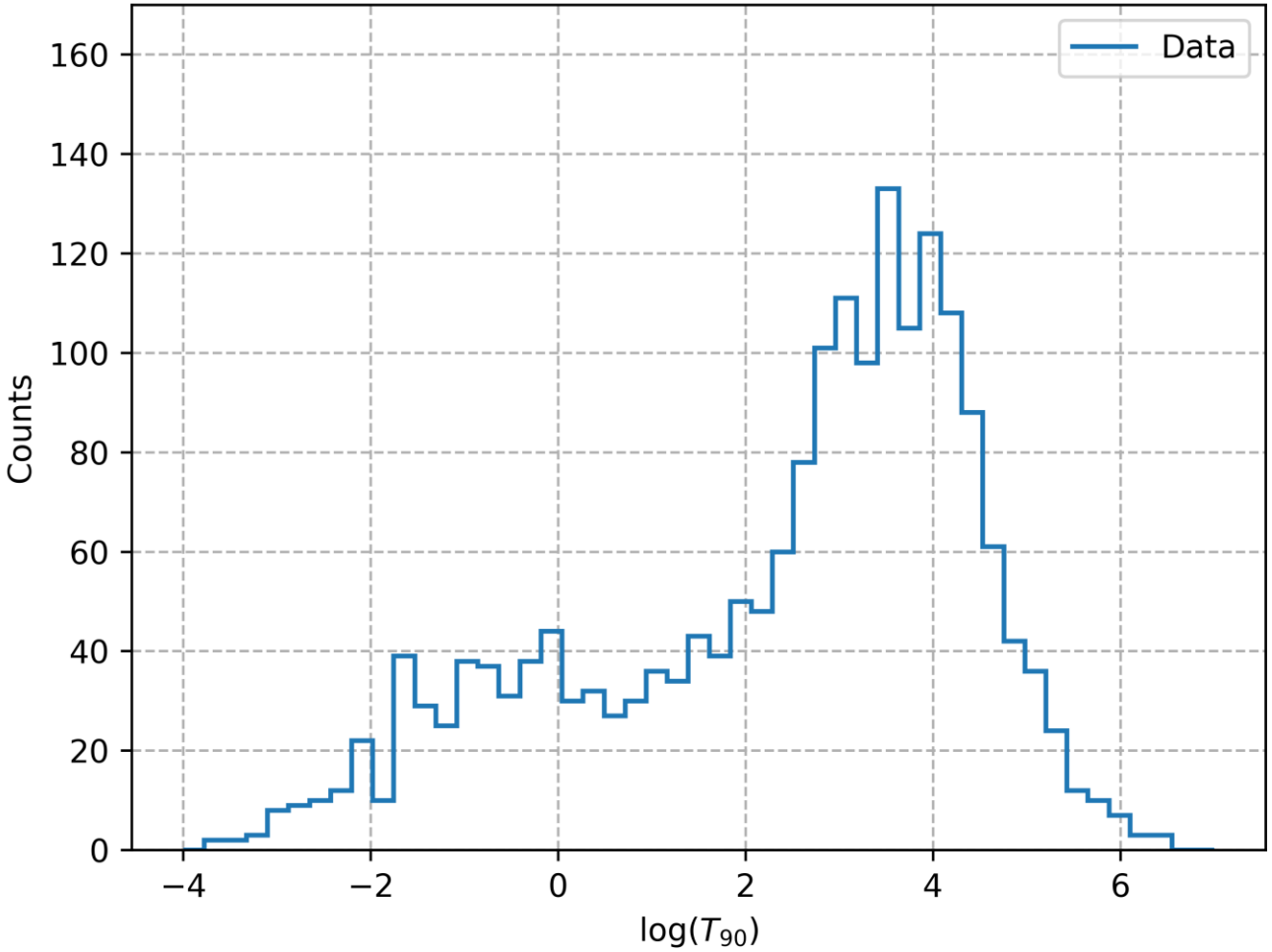
Dataset: 1932 burst seen by BATSE with the recorded $\log(T_{90})$, associated error and $\log(HR)$

Histogram on $\log(T_{90})$

Smallest value: ~ -3.7
Biggest value: ~ 6.5

50 bins from -4 to 7

Mean error is: 0.174
Median error is: 0.0919
A bin is large 0.224



Classification on $\log(T_{90})$: model

The model for the classification is the weighted sum of two log normal distribution

$$p(\log(T_{90})) = w\, N(\log(T_{90})\,|\mu_1\sigma_1) + (1-w)\, N(\log(T_{90})\,|\mu_2\sigma_2)$$

So we have 5 parameters to estimate, we will call them $\theta \equiv [w,\, \mu_1,\, \sigma_1,\, \mu_2,\, \sigma_2]$

Classification on $\log(T_{90})$: Bayes theorem

To estimate the parameters we will use Bayes theorem, define the propositions

- θ = «The set of parameters is θ»
- D = «The dataset we are analysing»
- H = «Model of the two weighted sum log normal»
- I = «All the background information on GRBs»

$$P(\theta|HDI) = P(\theta|HI)\frac{P(D|\theta HI)}{P(D|HI)}$$

Or in the log form:

$$\log P\,(\theta|HDI) = \log P\,(\theta|HI) + \log P\,(D|\theta HI) - \log P\,(D|HI)$$

Log prior — Log likelihood — Normalization costant

Classification on $\log(T_{90})$: Prior

$P(\theta|HI)$ Probability for a set of parameters given the model and background. Outside of the bounds is zero, inside?

For the weight and the means we chose a uniform prior

$$0 < w < 1$$

$$-4 < \mu_i < 7$$

$$Prior \propto 1$$

For the sigmas we chose a Jeffrey's prior because this are scale parameters of the problem

$$0.01 < \sigma_i < 3$$

$$Prior \propto 1/\sigma$$

Classification on $\log(T_{90})$: Likelihood

$P(D|\theta HI)$ Probability of seeing the dataset given a set of parameters, the model and background

Having the model, the parameters and the bins we can compute the predicted count in a bin by our model

$$\lambda_b = N \cdot p(\log(T_{90})_b\,|\theta) \cdot db$$

The probability of having the count c_b in a bin given the predicted count λ_b is the Poisson's distribution

$$P(c_b|\lambda_b) = \frac{1}{c_b!}\lambda_b^{c_b} e^{-\lambda_b}$$

The likelihood is the sum over all the bins!

$$P(D|\theta HI) = \sum_b P(c_b|\theta HI)$$

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

$q(\theta)$ is the proposal distribution

We chose a 5D multivariate gaussian but how we chose his covariance matrix?

- «Blind» run: we use a diagonal covariance matrix with elements equals to 5% of the bound
- «Not Blind» run: we use a covariance matrix estimated after a previous run

Base idea

- Sample a θ_t from $q(\theta_0)$

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

Acceptance rule:

$$\frac{p(\theta_t)}{p(\theta_0)} \cdot \frac{q(\theta_t|\theta_0)}{q(\theta_0|\theta_t)} \geq a$$

Posterior values

Proposal probability ratio
ours is symmetric so it's one

Random number
from [0,1]

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

If true move to $\theta_0 = \theta_t$

If false stay at θ_0

In this way we will have the θ with higher posterior more times so more probable!

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

We make a first «blind» run of the algorithm with 10k steps to estimate the covariance

We then make the final run with 100k steps

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

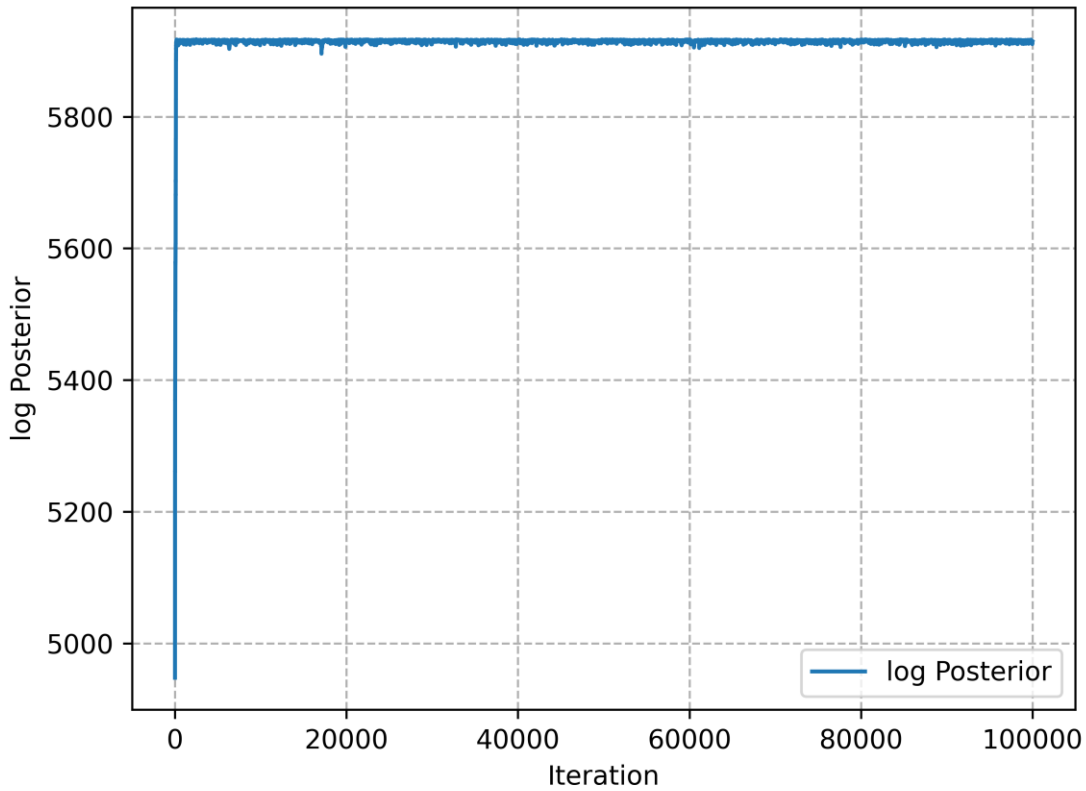
If true move to $\theta_0 = \theta_t$

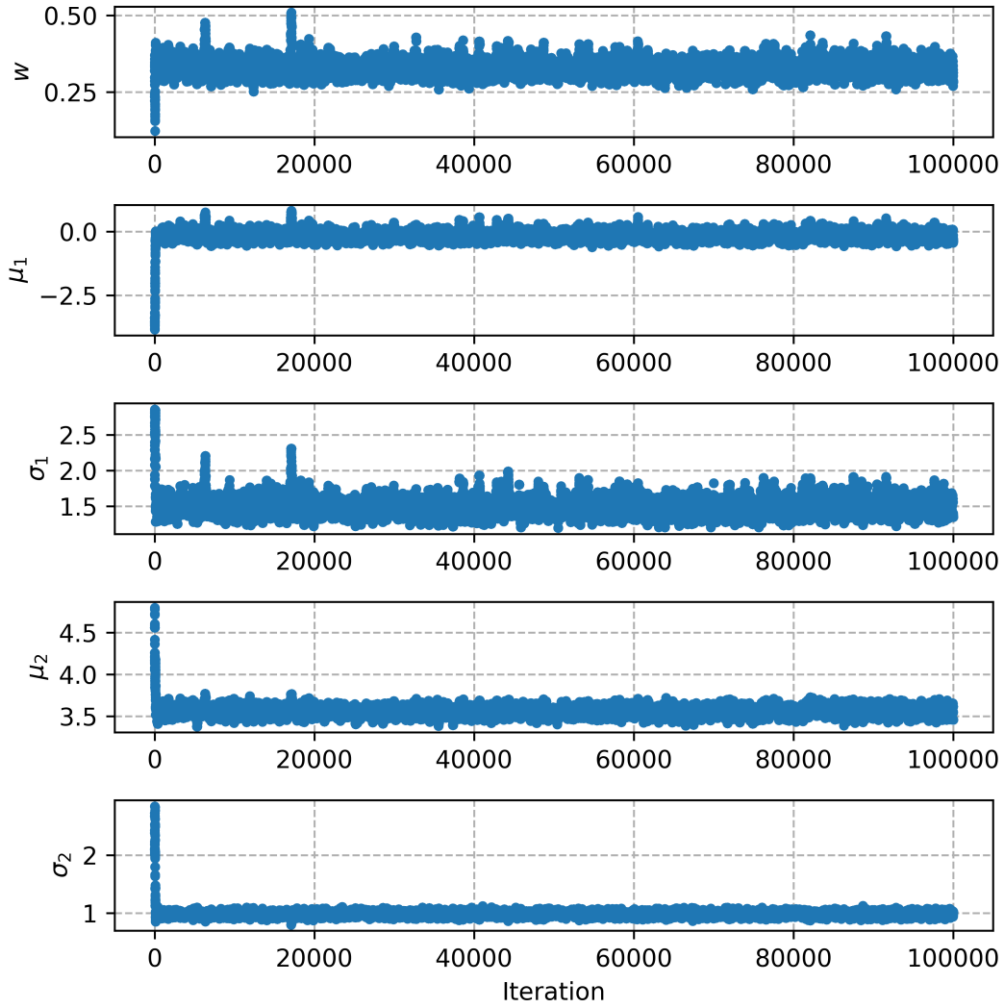
If false stay at θ_0

In this way we will have the θ with higher posterior more times so more probable!

Classification on $\log(T_{90})$: traces

We see that some steps are needed to find the peak → Burn-in: discard n points to have samples of the peak

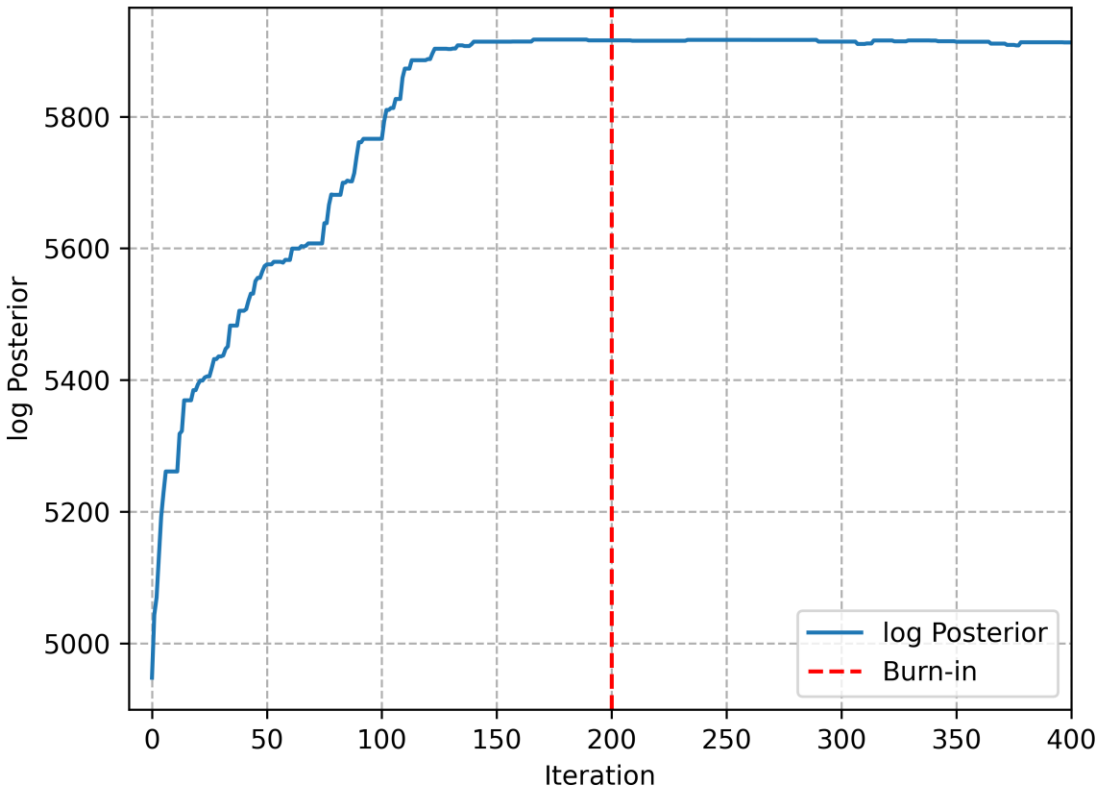


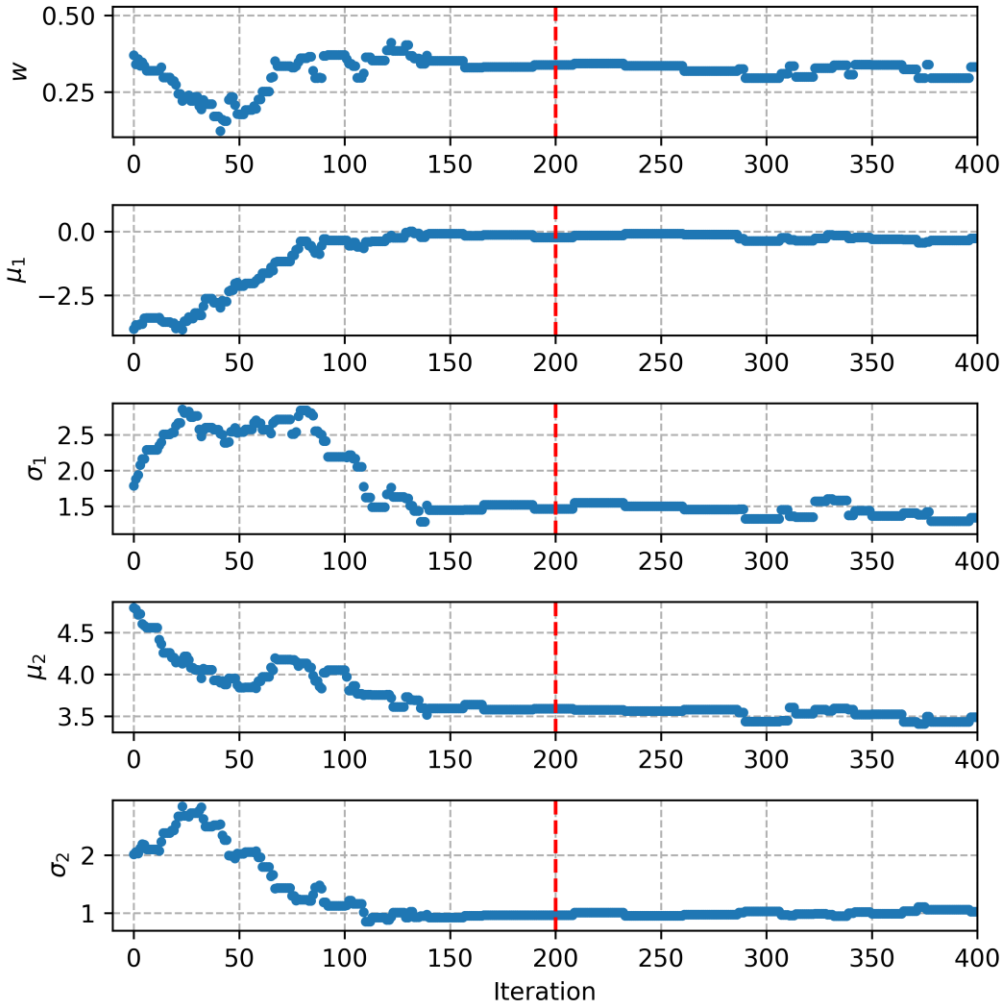


Classification on $\log(T_{90})$: traces

We see that some steps are needed to find the peak → Burn-in: discard n points to have samples of the peak

Burn-in = 200

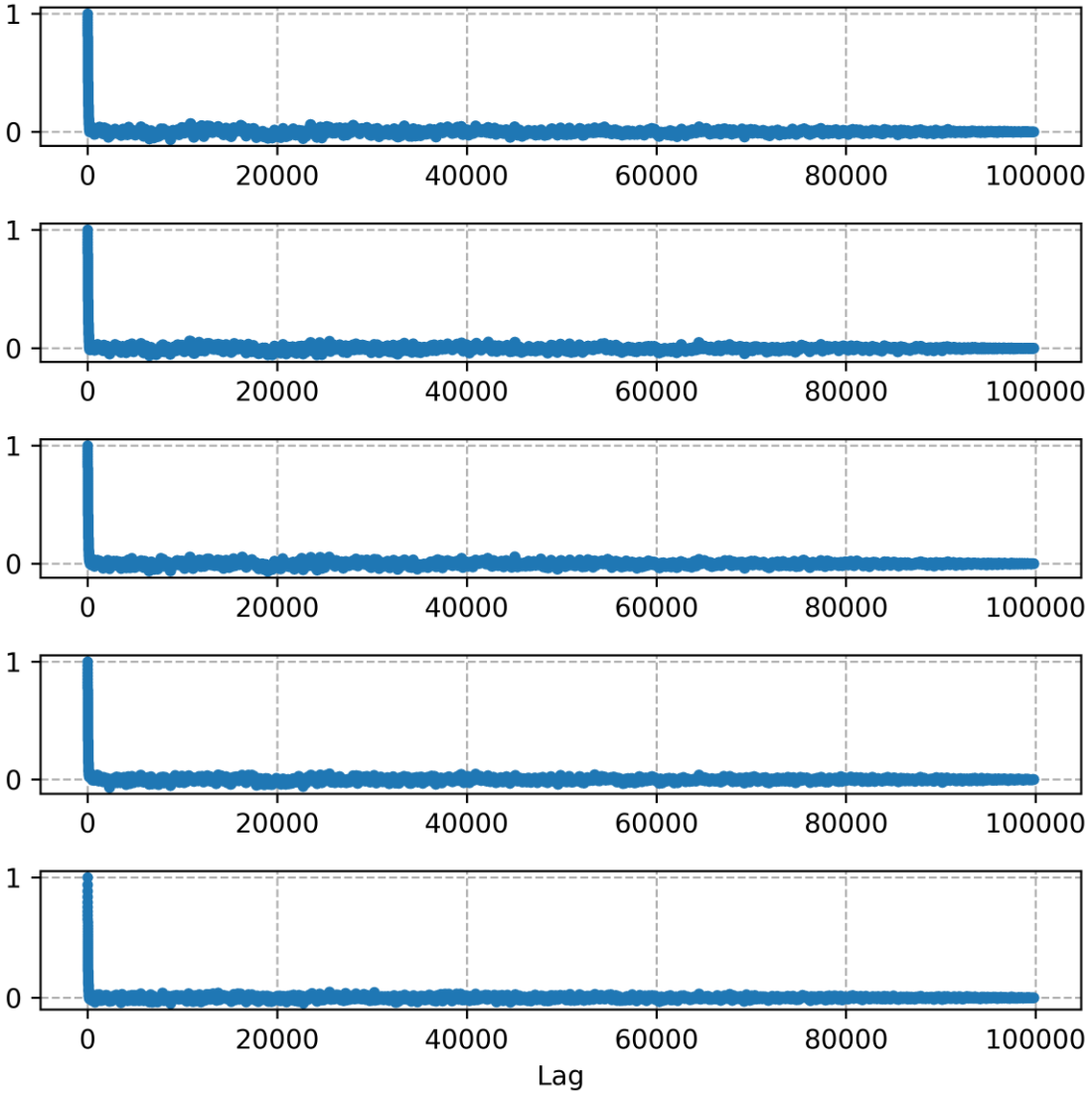




Classification on $\log(T_{90})$: autocorrelation

We look at the autocorrelation for the parameters and we see that for small lag the samples are correlated

Thinning: take a point every k to have not correlated samples



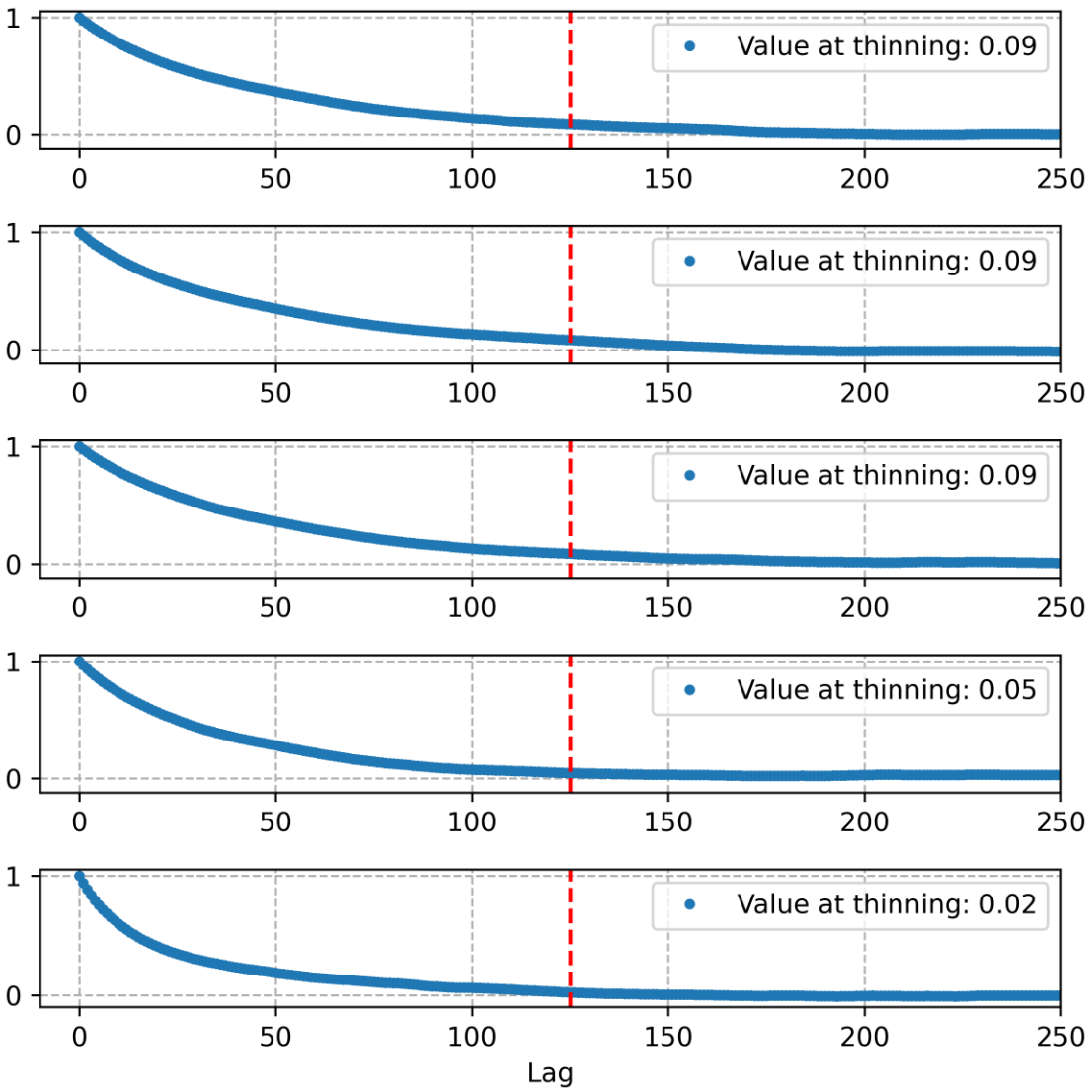
Classification on $\log(T_{90})$: autocorrelation

We look at the autocorrelation for the parameters and we see that for small lag the samples are correlated

Thinning: take a point every k to have not correlated samples

Thinning = 125

Remaining samples = 799



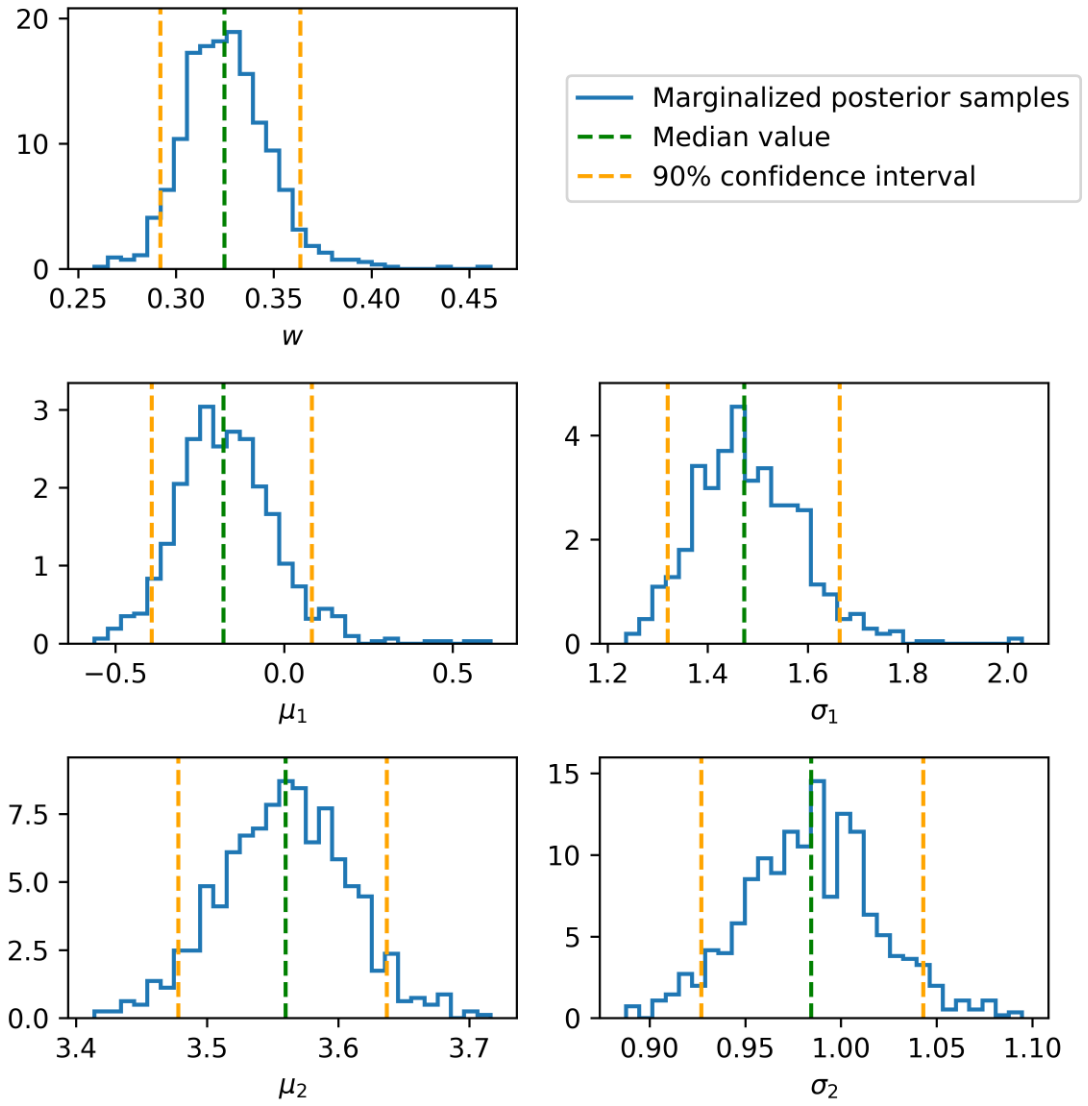
Classification on $\log(T_{90})$: marginals

We make the histograms for each parameter
We find the median value and the interval corresponding to 90% level of confidence

w
$0.325^{+0.039}_{-0.033}$

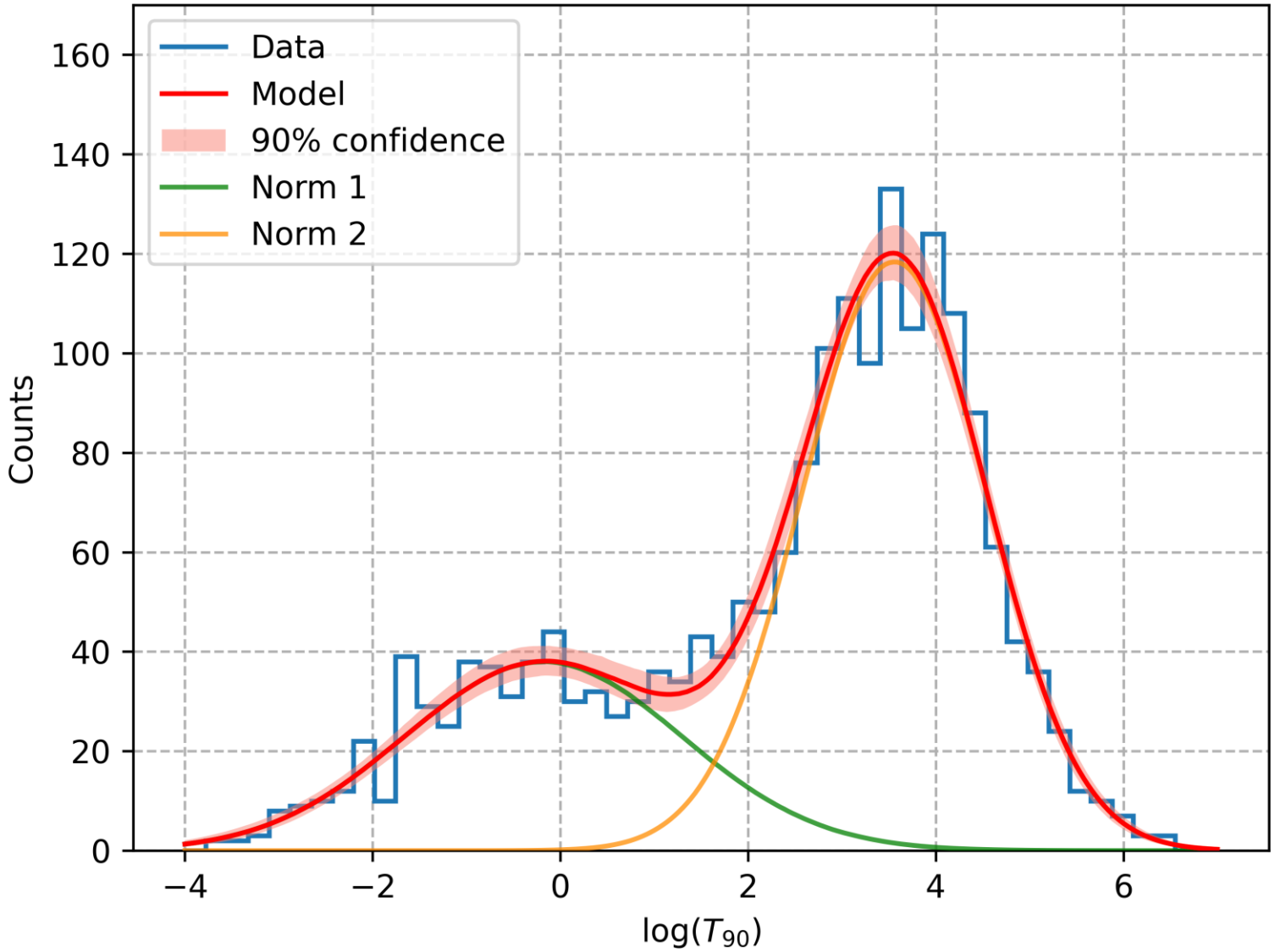
μ_1	σ_1
$-0.18^{+0.26}_{-0.21}$	$1.47^{+0.19}_{-0.15}$

μ_2	σ_2
$3.560^{+0.077}_{-0.082}$	$0.984^{+0.058}_{-0.057}$



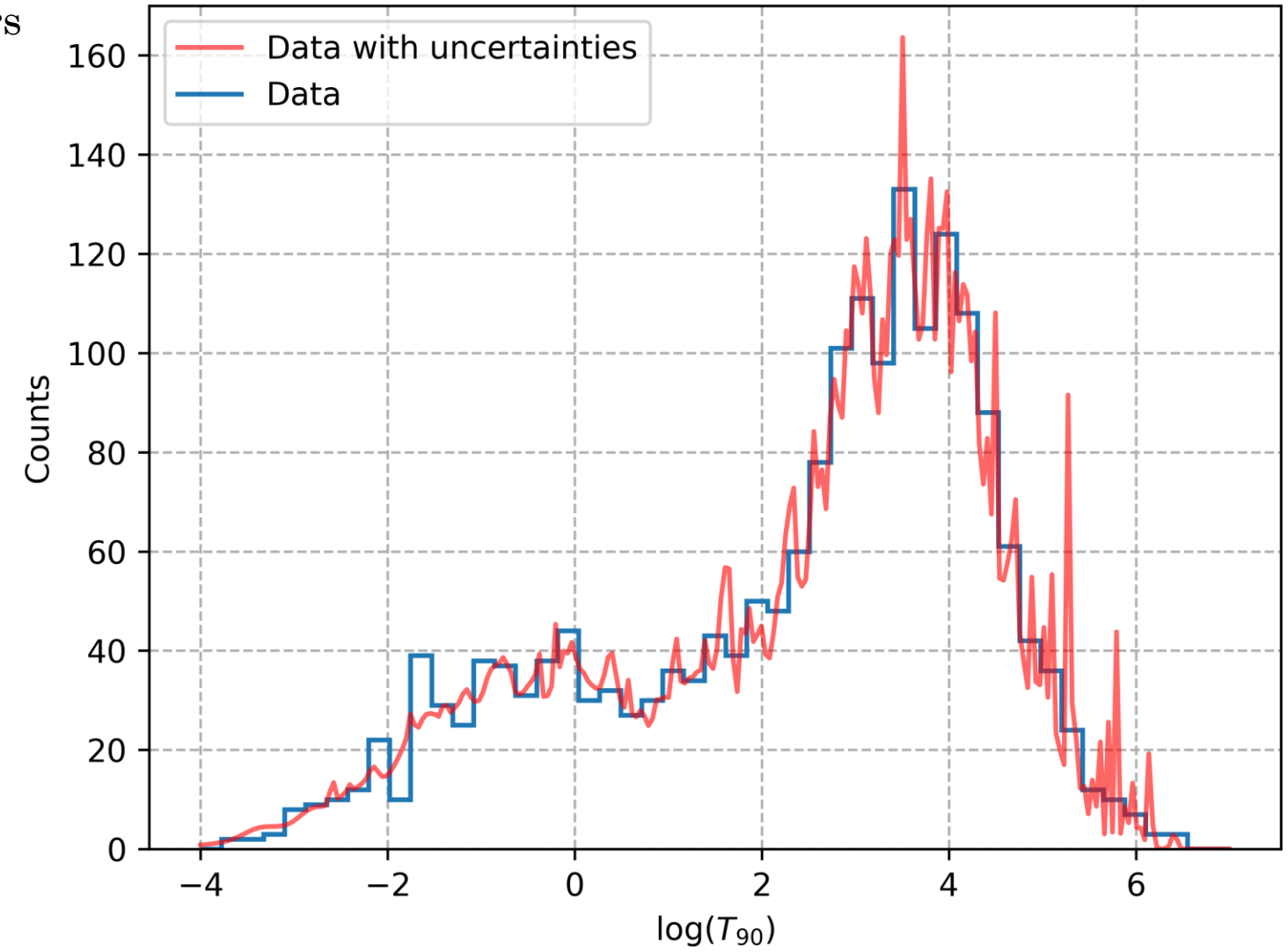
Classification on $\log(T_{90})$: final model

The final model, the median of the models computed on each sample



Classification on $\log(T_{90})$ with errors

Now consider the errors as gaussian errors on each measure



To visualize the data we sum all the gaussian $N(\log(T_{90})_i, \sigma_i)$, we normalize and scale to the binned data

Classification on $\log(T_{90})$ with errors

- Same model
- Same prior
- Same Metropolis Hasting Algorithm
- Different likelihood!

Given the uncertainties we can't histogram the data and we have to consider each measure as a realization from a gaussian with unknown mean

Calling $\log(T_{90}) = x$

$$\mathcal{L}_i = \int p(x_t|\theta) \cdot N(x_i|x_t\sigma_i)dx_t$$

Probability from the model of having x_t

Probability of measuring x_i given the mean x_t

Classification on $\log(T_{90})$ with errors

- Same model
- Same prior
- Same Metropolis Hasting Algorithm
- Different likelihood!

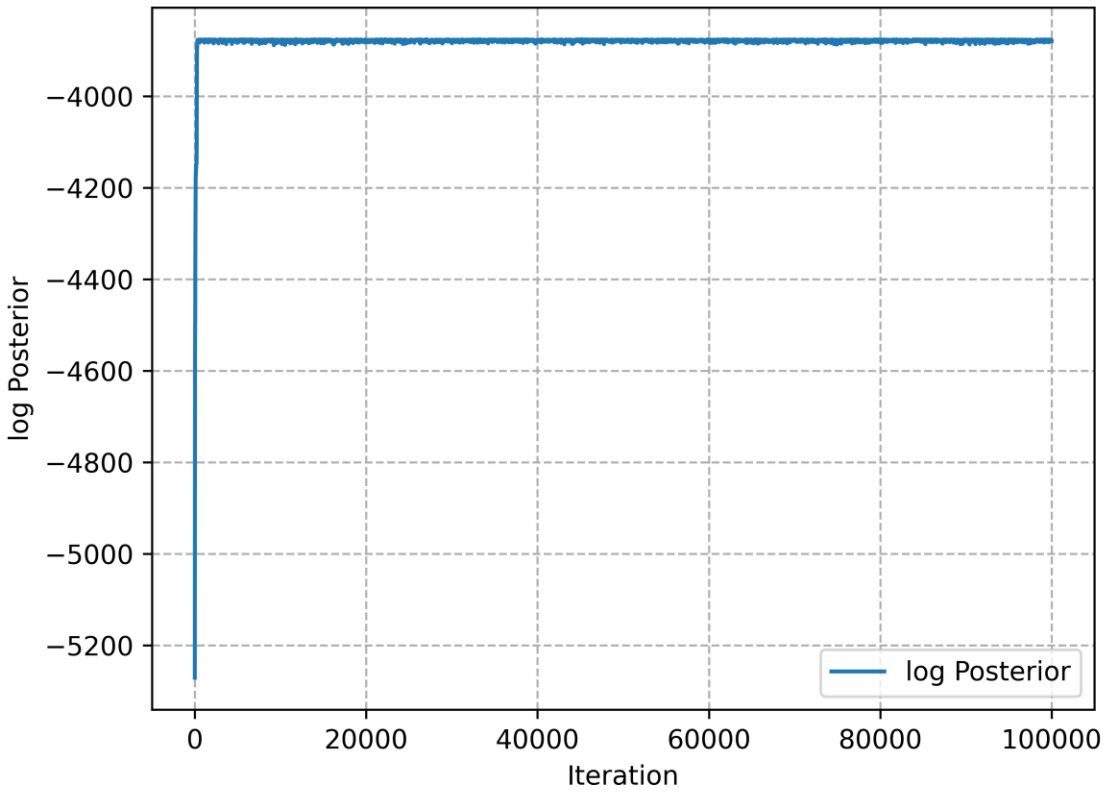
$$\mathcal{L}_i = \int p(x_t|\theta) \cdot N(x_i|x_t\sigma_i)dx_t$$

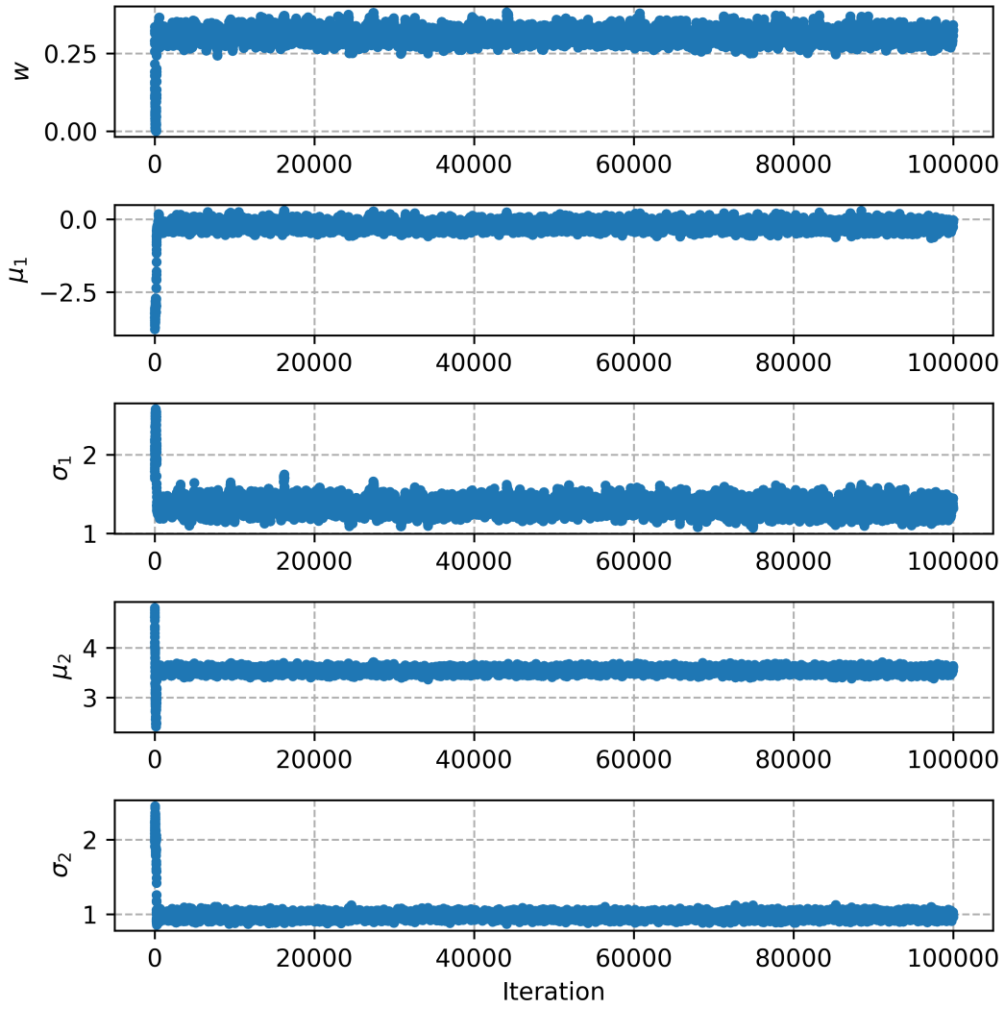
Fortunately our model is formed by gaussians so this integral is simply

$$\mathcal{L}_i = w \cdot N\left(x_i \middle| \mu_1, \sqrt{\sigma_1^2 + \sigma_i^2}\right) + (1-w) \cdot N\left(x_i \middle| \mu_1, \sqrt{\sigma_2^2 + \sigma_i^2}\right)$$

And the final likelihood will be the sum over the number of measure

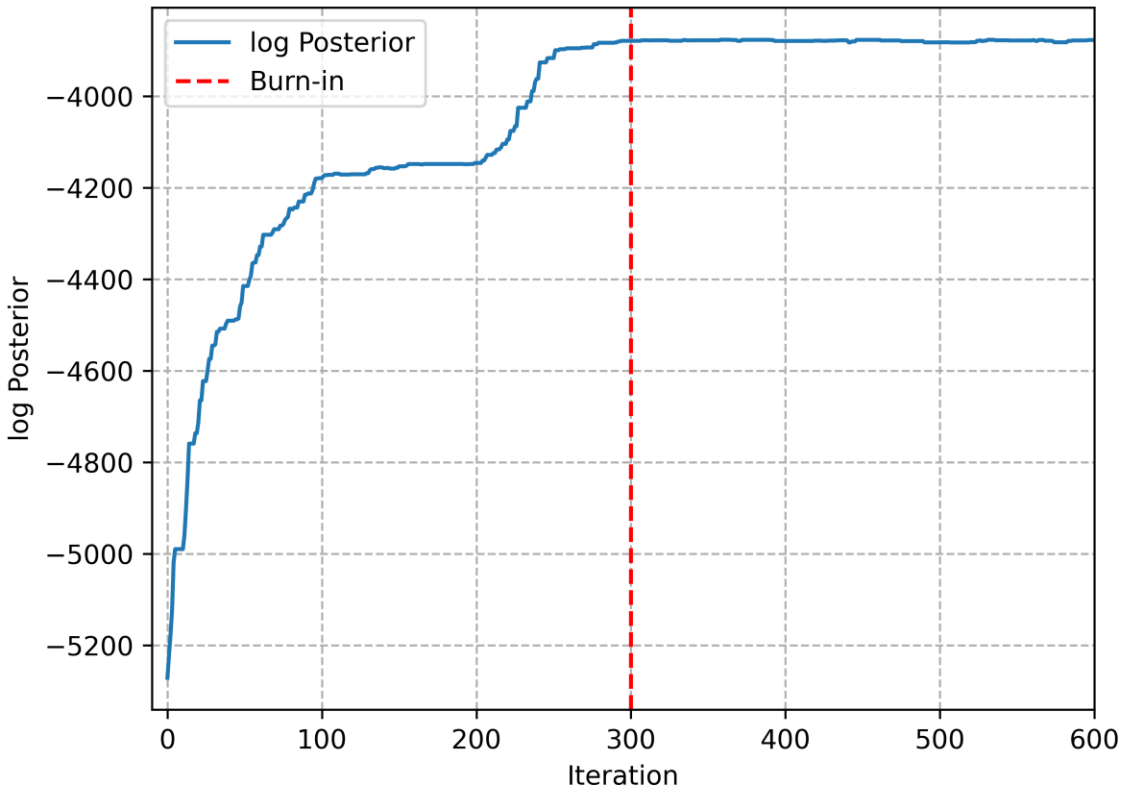
Classification on $\log(T_{90})$ with errors

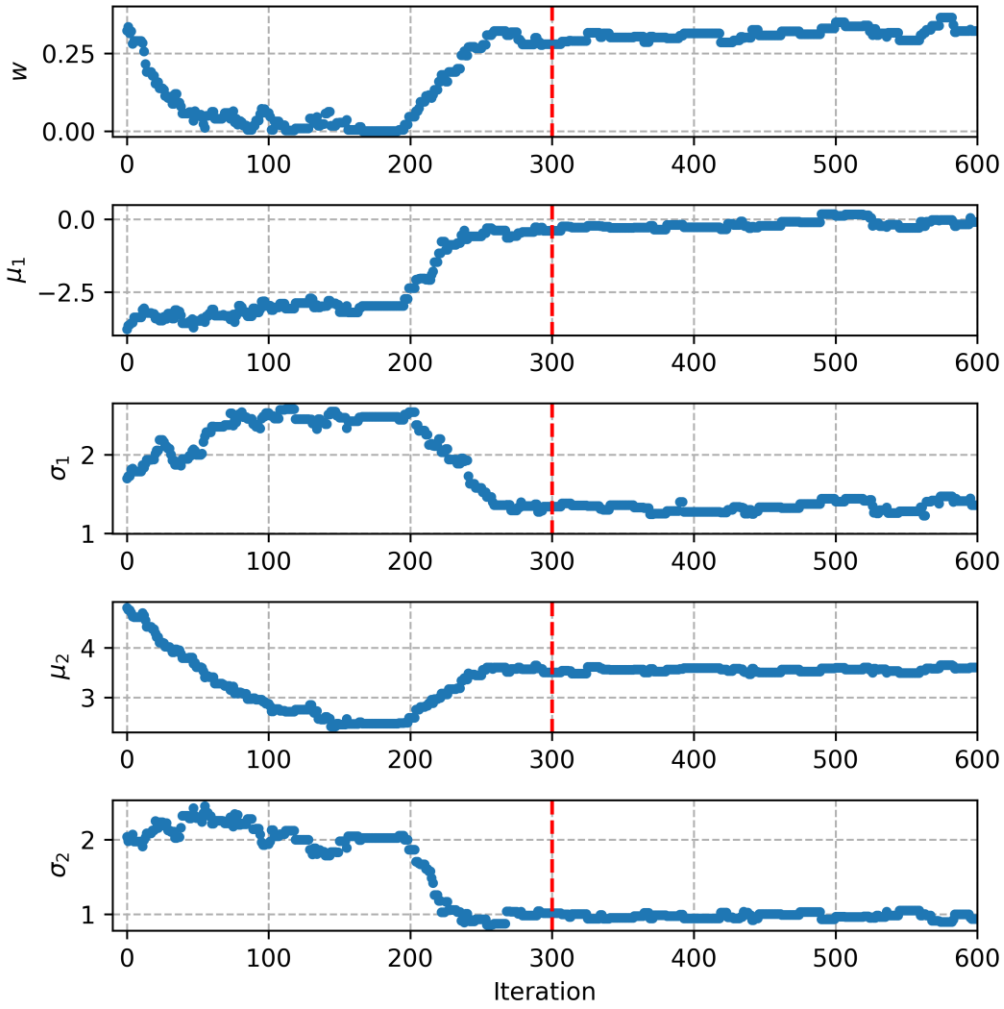




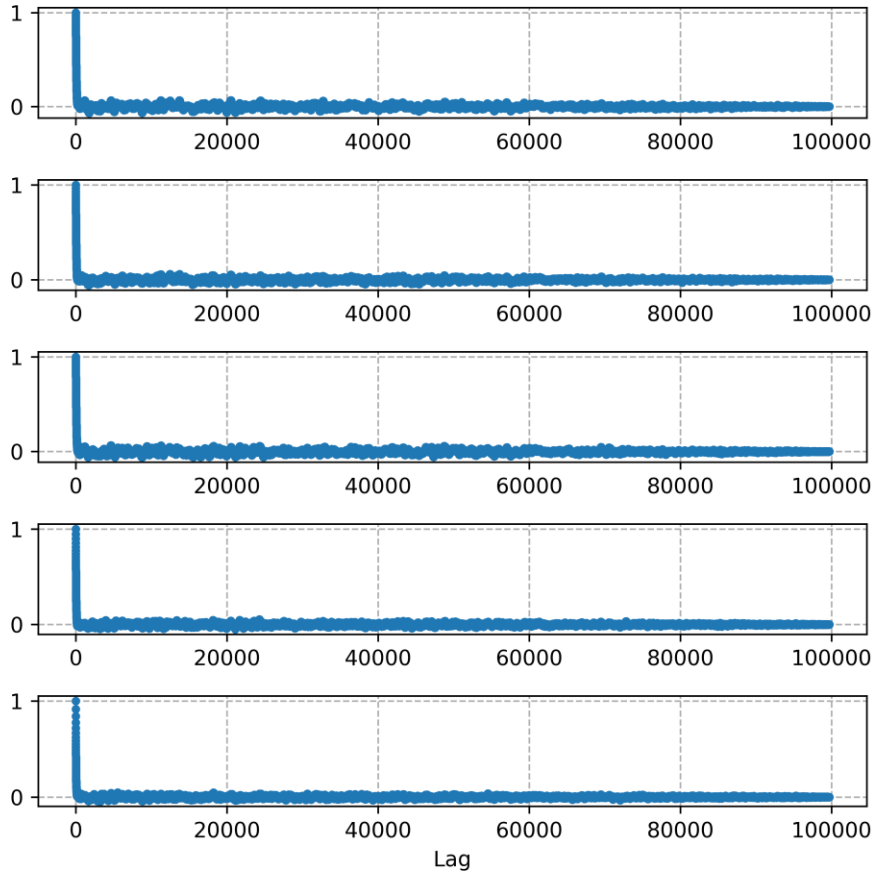
Classification on $\log(T_{90})$ with errors

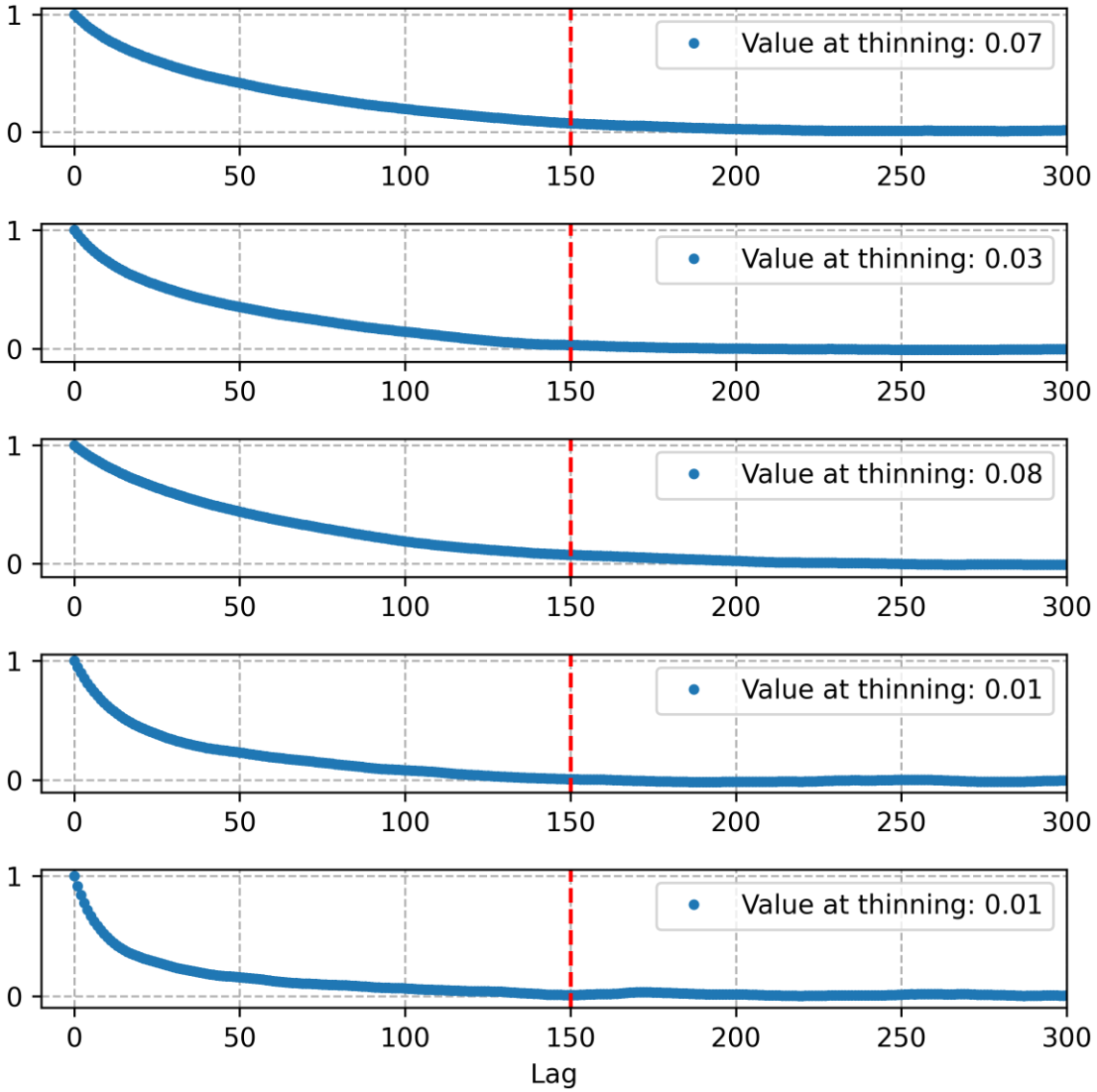
Burn-in = 300





Classification on $\log(T_{90})$ with errors





Thinning = 150

Remaining samples = 665

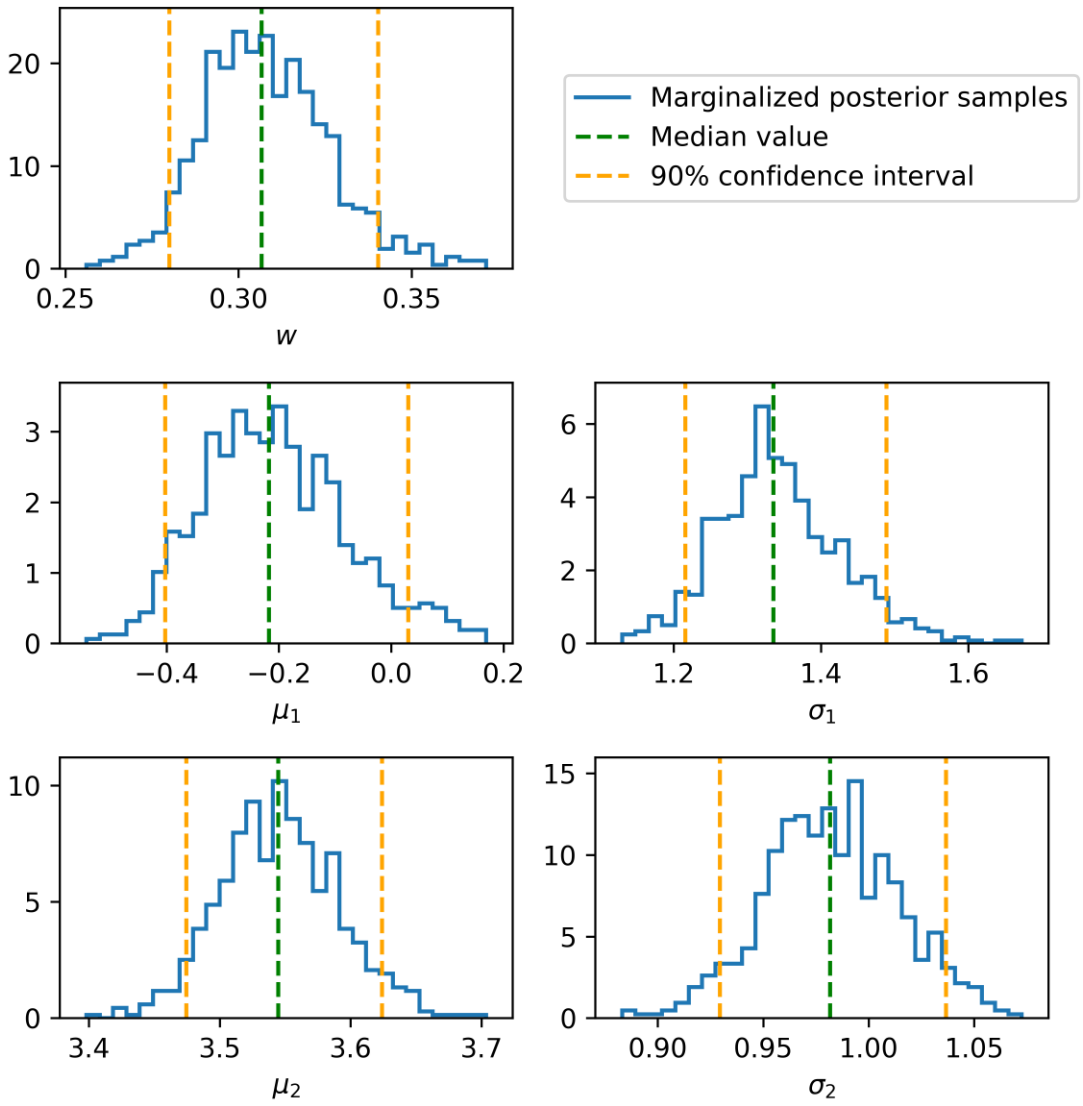
Classification on $\log(T_{90})$ with errors

We make the histograms for each parameter
We find the median value and the interval corresponding to 90% level of confidence

w
$0.307^{+0.034}_{-0.027}$

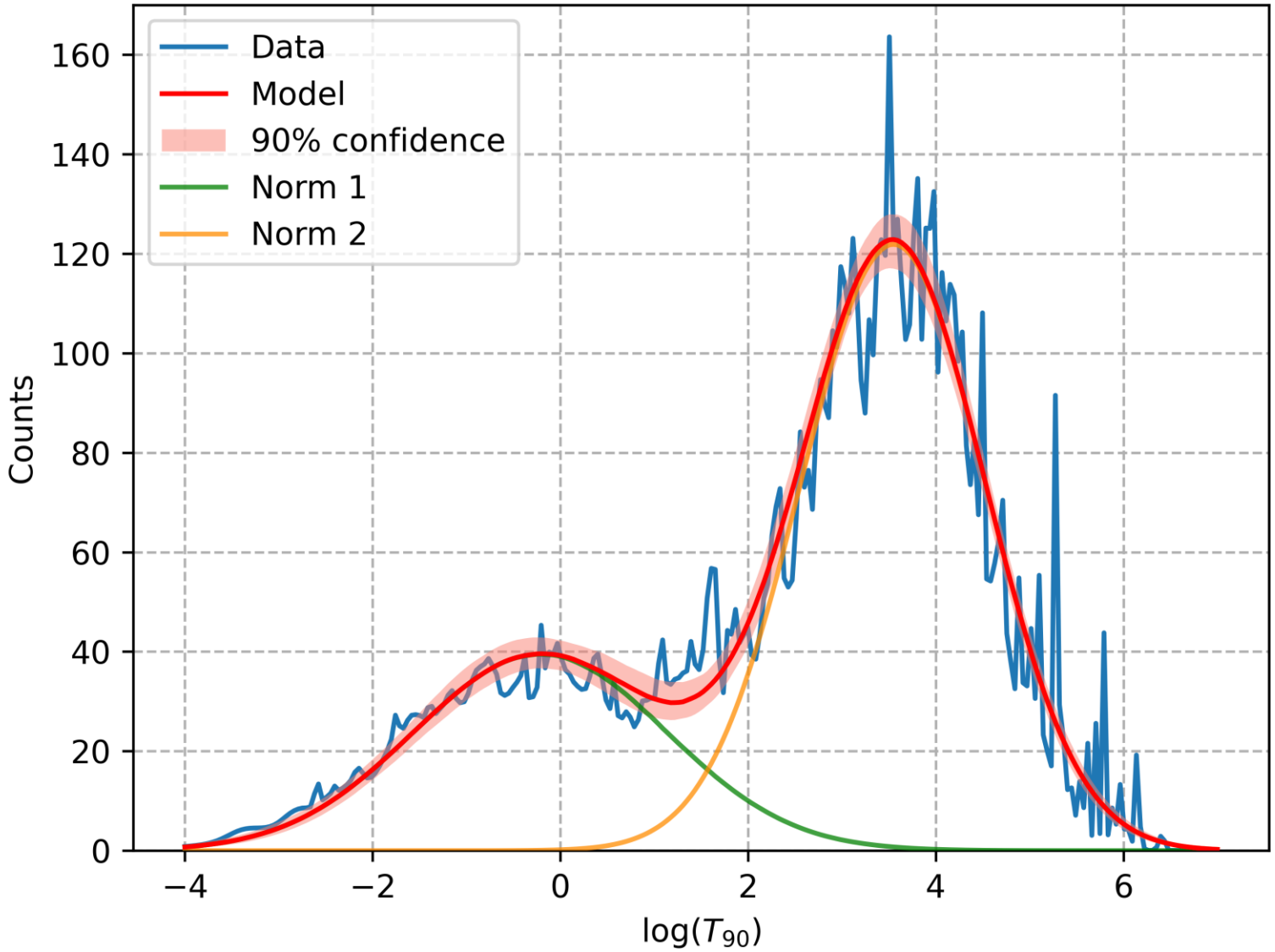
μ_1	σ_1
$-0.22^{+0.25}_{-0.18}$	$1.34^{+0.15}_{-0.12}$

μ_2	σ_2
$3.544^{+0.079}_{-0.070}$	$0.982^{+0.055}_{-0.052}$

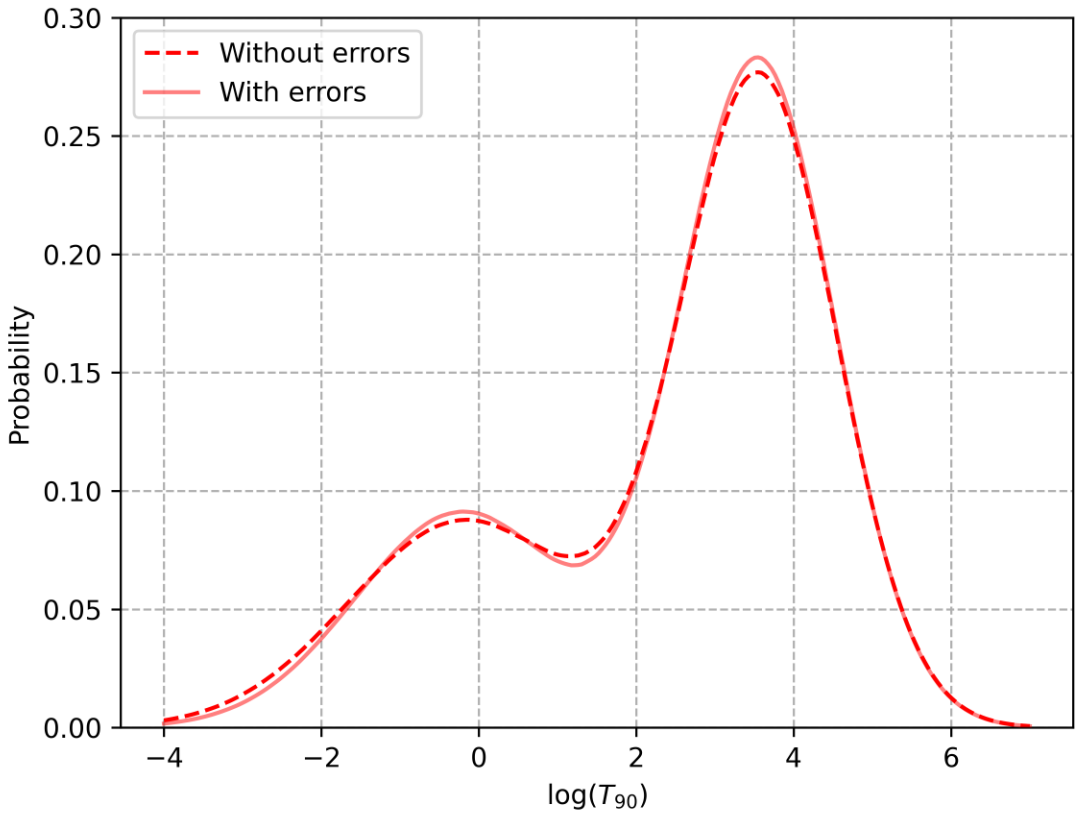


Classification on $\log(T_{90})$ with errors

The final model, the median of the models computed on each sample

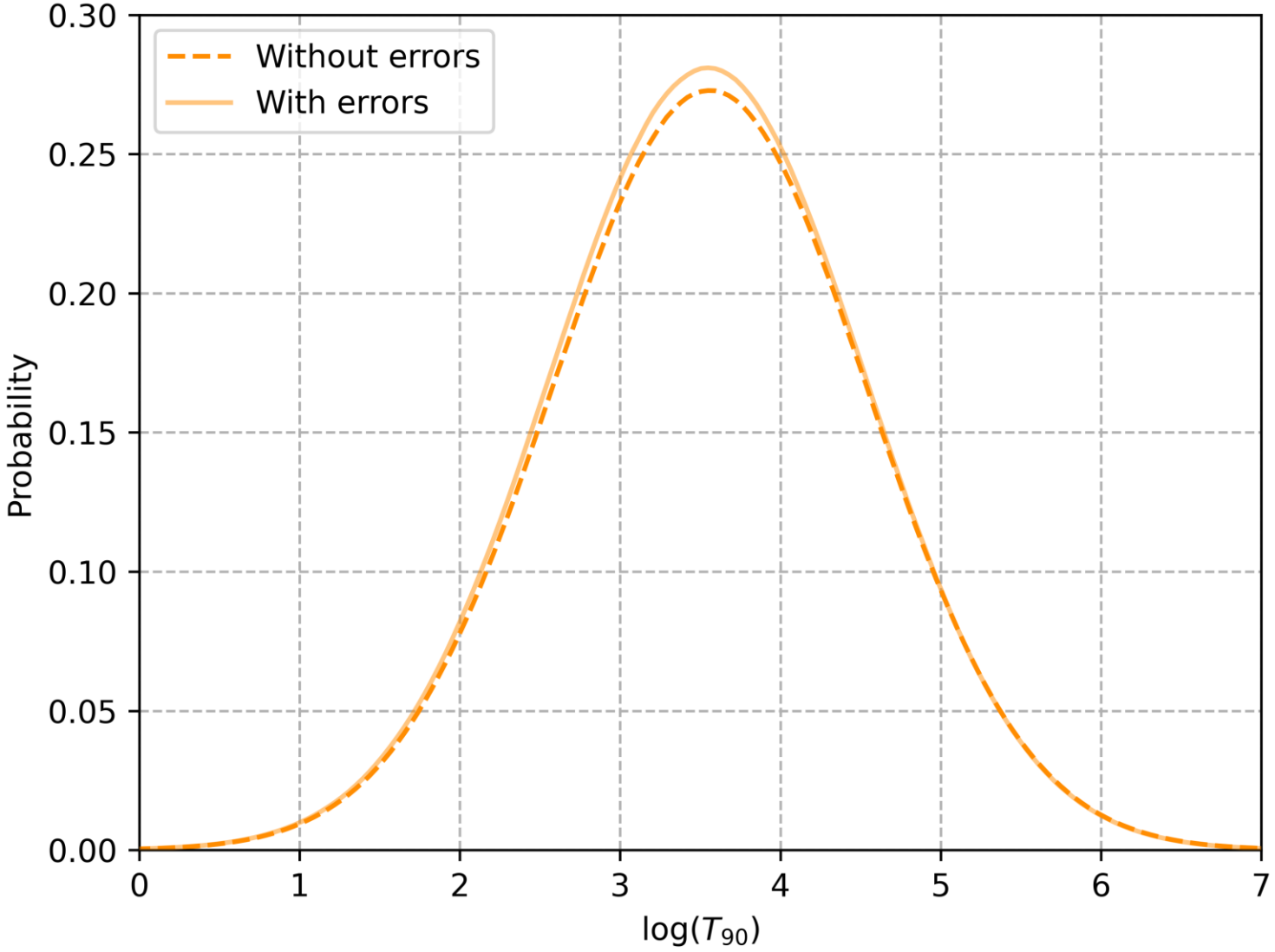


Classification on $\log(T_{90})$: differences

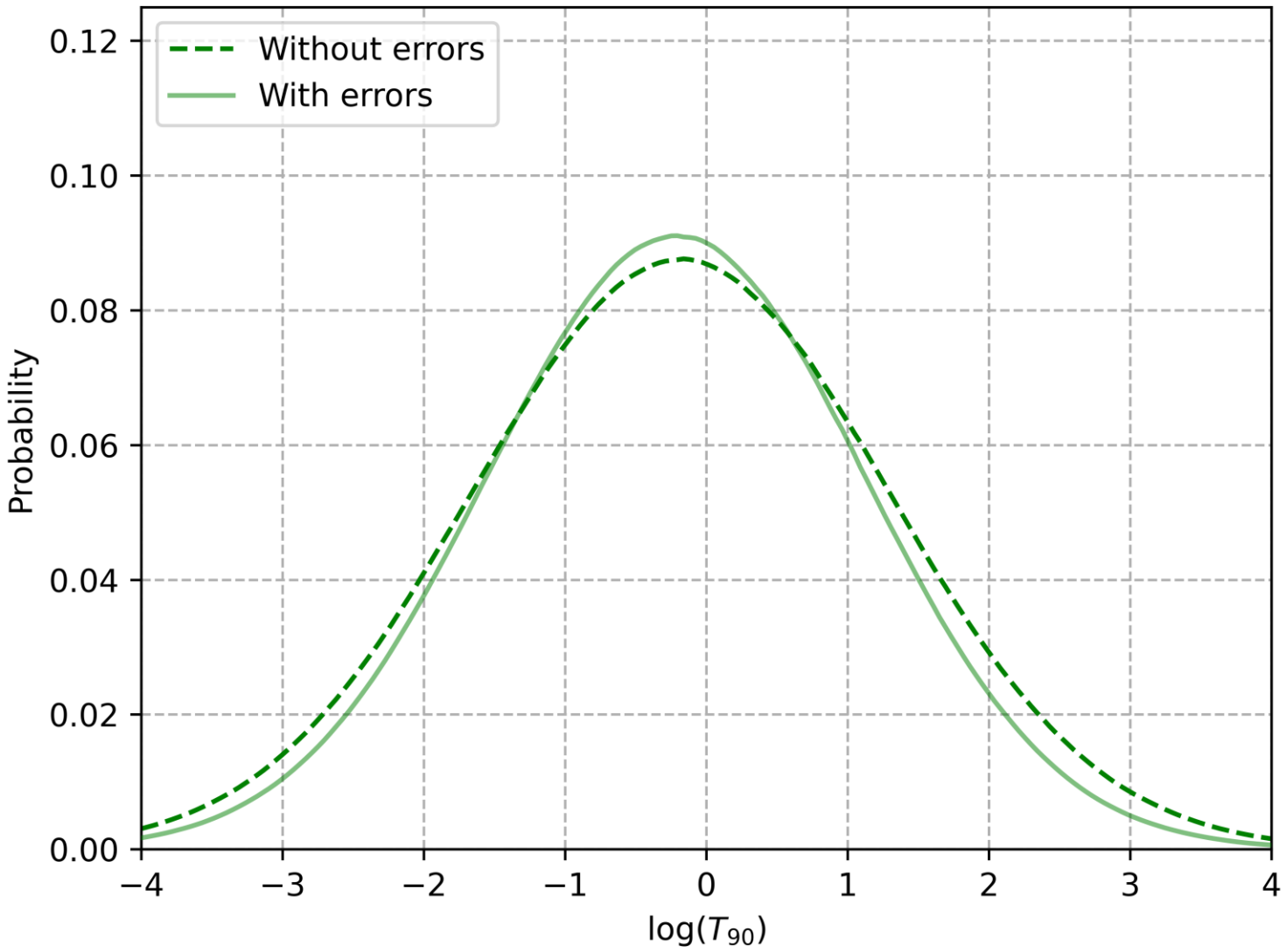


	w	μ_1	σ_1	μ_2	σ_2
Without errors	$0.325^{+0.039}_{-0.033}$	$-0.18^{+0.26}_{-0.21}$	$1.47^{+0.19}_{-0.15}$	$3.560^{+0.077}_{-0.082}$	$0.984^{+0.058}_{-0.057}$
With errors	$0.307^{+0.034}_{-0.027}$	$-0.22^{+0.25}_{-0.18}$	$1.34^{+0.15}_{-0.12}$	$3.544^{+0.079}_{-0.070}$	$0.982^{+0.055}_{-0.052}$

Classification on $\log(T_{90})$: differences



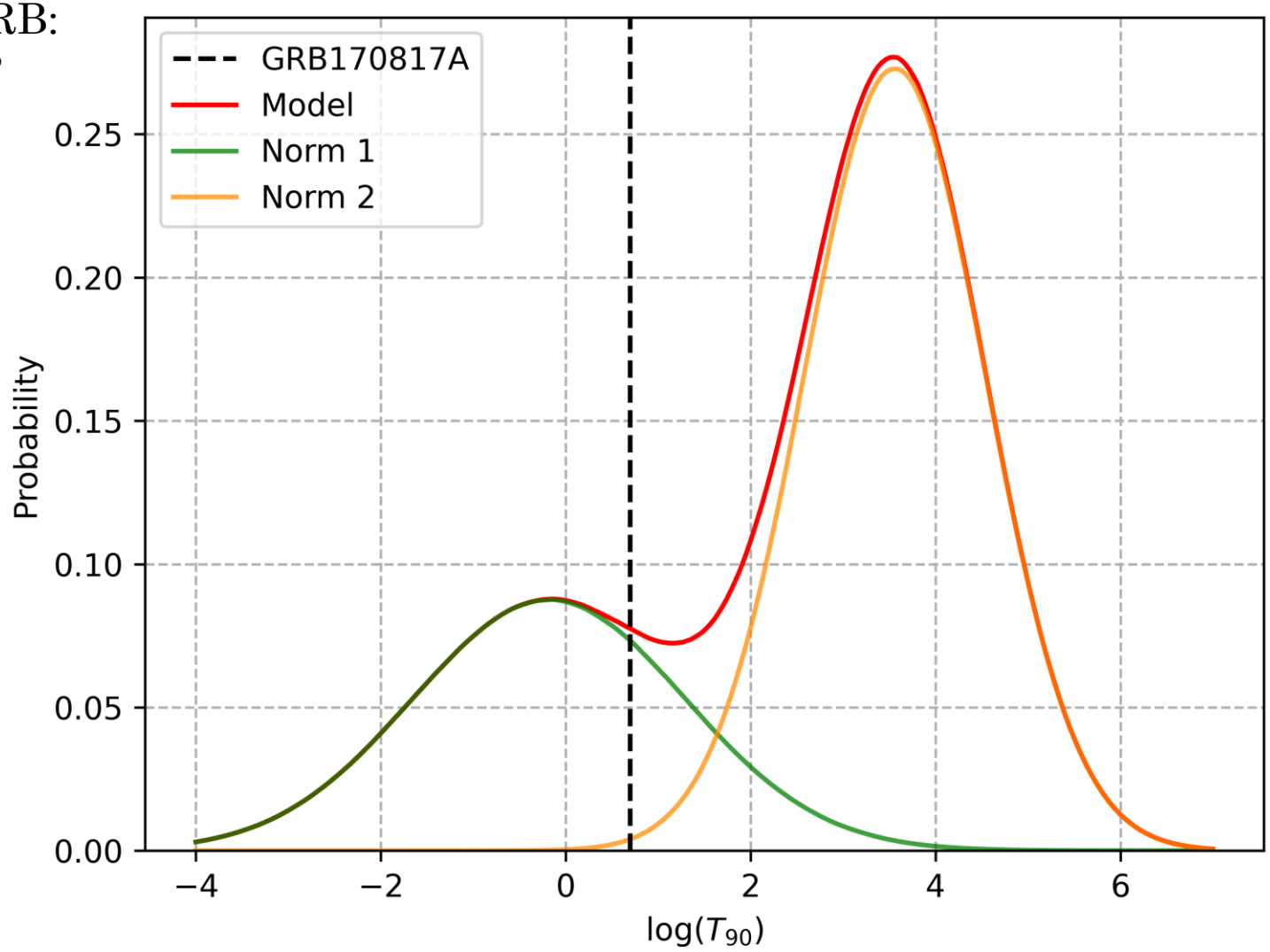
Classification on $\log(T_{90})$: differences



Classifying a GRB

We now look at the problem of classifying a single GRB: it's a short or long GRB?

Case study of GRB170817A, $T_{90} = 2.0$ s



Classifying a GRB

We will once again use Bayes theorem, start defining the propositions

- T = «The GRB has this T_{90}»
- S = «The GRB is a short GRB»
- L = «The GRB is a long GRB»
- I = «Model we found (without errors) and all the background information on GRBs»

So for the short:

$$P(S|TI) = P(S|I) \cdot \frac{P(T|SI)}{P(T|I)}$$

And the long:

$$P(L|TI) = P(L|I) \cdot \frac{P(T|LI)}{P(T|I)}$$

Classifying a GRB

Priors on the two classes: probability that is short/long given the model

$$P(S|I) = w$$

$$P(L|I) = 1 - w$$

It's the weight of the two components

As expected they sum to 1, they are exhaustive proposition

Classifying a GRB

Priors on the two classes: probability that is short/long given the model

$$P(S|I) = w$$

$$P(L|I) = 1 - w$$

It's the weight of the two components

As expected they sum to 1, they are exhaustive proposition

Likelihood on the two classes: probability that is short/long given the model and a T_{90}

$$P(S|TI) = N(\log(T_{90})\,|\mu_1\ \sigma_1)$$

$$P(L|TI) = N(\log(T_{90})\,|\mu_2\ \sigma_2)$$

It's the two normal distributions evaluated at the given T_{90}

Classifying a GRB

Normalization constant for the two classes: probability of having a T_{90} given the model

$$P(T|I) = p(\log(T_{90})\,|\theta)$$

It's the total model evaluated at the given T_{90}

Classifying a GRB

Normalization constant for the two classes: probability of having a T_{90} given the model

$$P(T|I) = p(\log(T_{90})\,|\theta)$$

It's the total model evaluated at the given T_{90}

Putting together this pieces we obtain for GRB170817A

$$P(S|TI) = 95\%$$

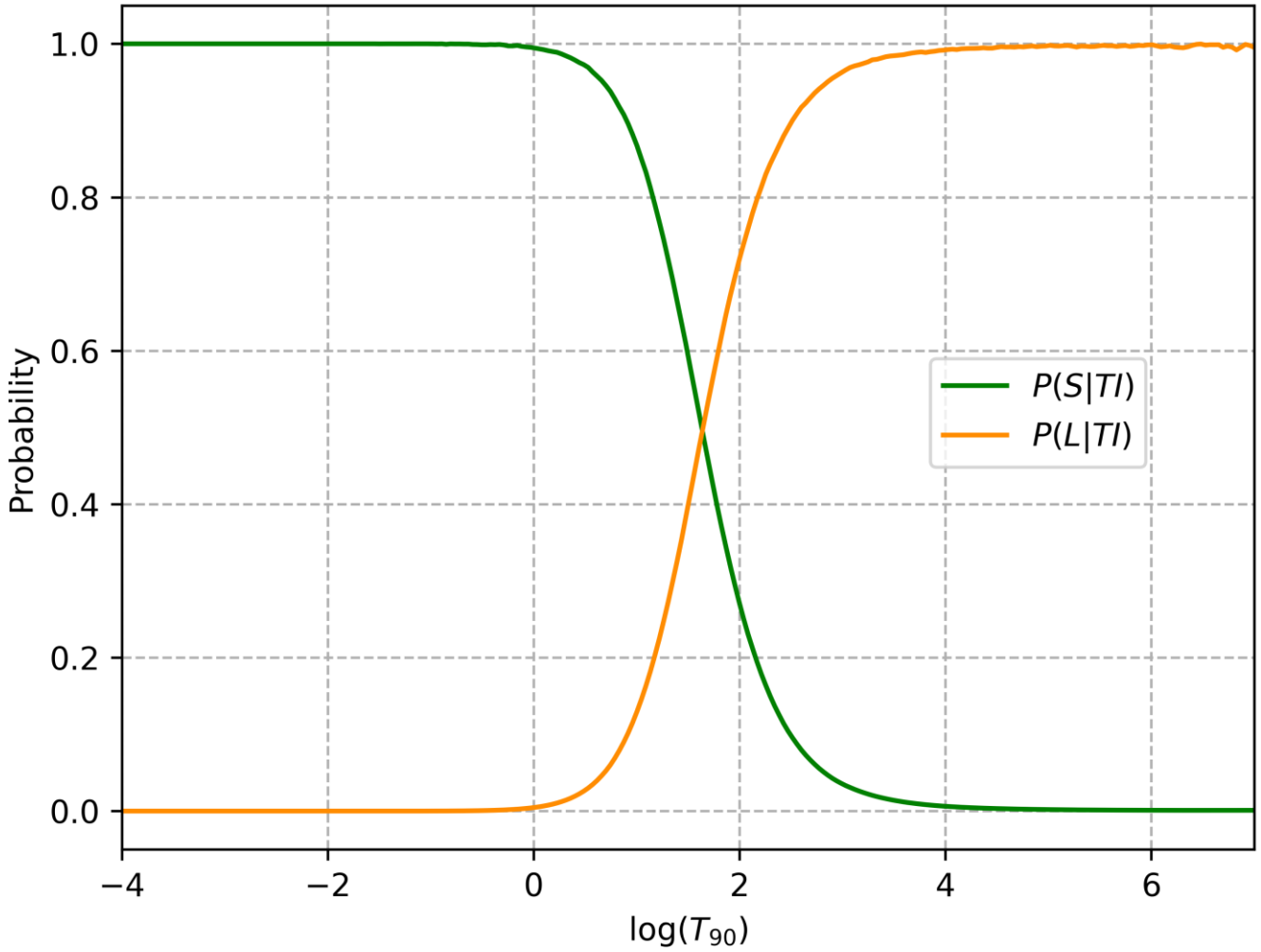
$$P(L|TI) = 5\%$$

So for our model it's a short GRB.

At which T_{90} we start to change class?

Classifying a GRB

At which T_{90} we start to change class?

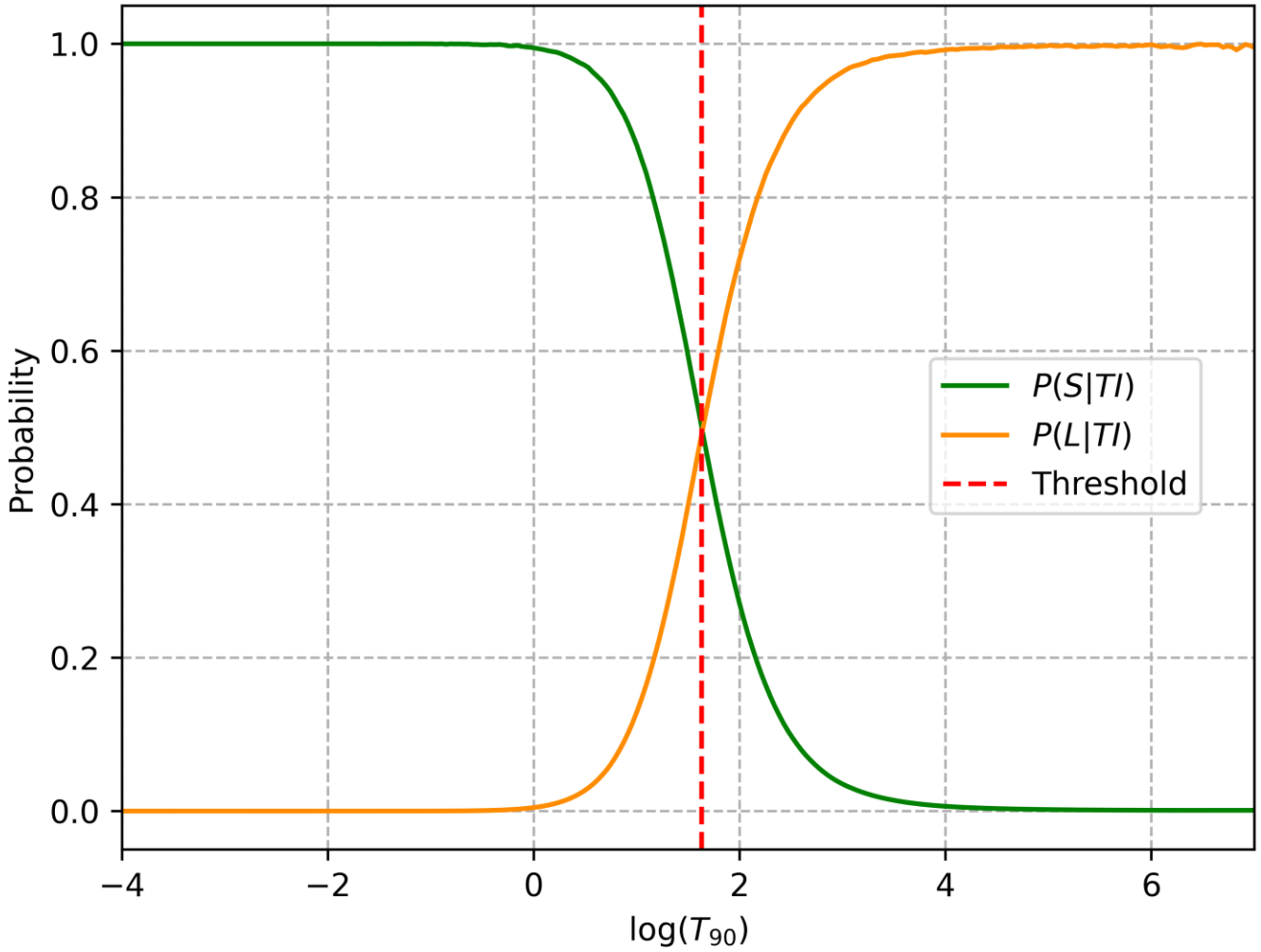


Classifying a GRB

At which T_{90} we start to change class?

We find the transition point

$\log(T_{90}) = 1.63 \Rightarrow T_{90} = 5.10$ s



Classifying a GRB

At which T_{90} we start to change class?

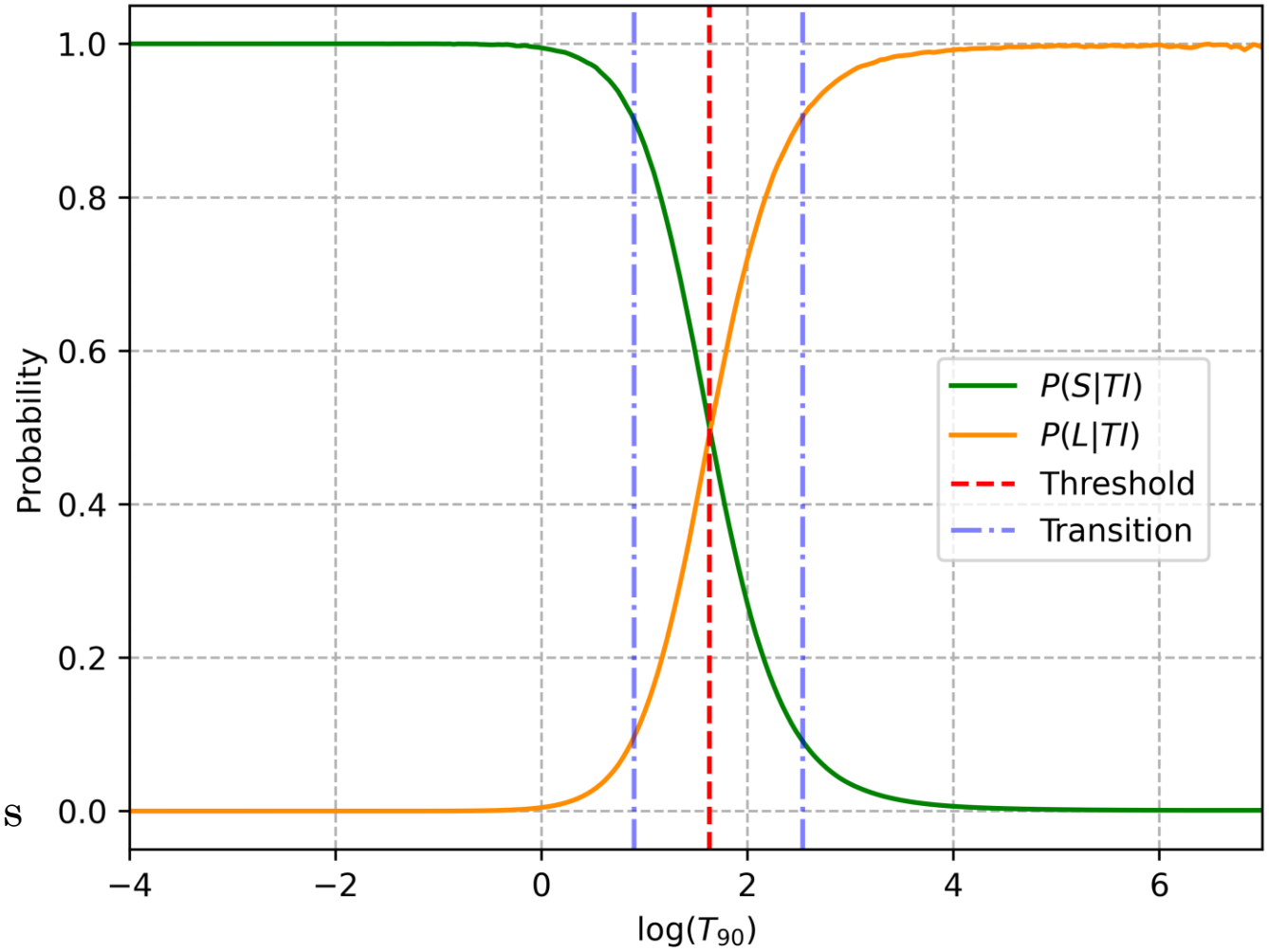
We find the transition point

$\log(T_{90}) = 1.63 \Rightarrow T_{90} = 5.10$ s

And the transition period

$\log(T_{90}) = 0.90 \Rightarrow T_{90} = 2.45$ s

$\log(T_{90}) = 2.54 \Rightarrow T_{90} = 12.62$ s



A third class of GRB?

In literature is proposed an intermediate class between the short and long

H_2 = «Model with two classes»

H_3 = «Model with three classes»

Which is favourite?

Bayes theorem on both:

$$P(H_i|DI) = P(H_i|I)\frac{P(D|H_iI)}{P(D|I)}$$

Odds ratio:

$$O_{23} = \frac{P(H_2|DI)}{P(H_3|DI)} = \frac{P(D|H_2I)}{P(D|H_3I)}$$

We are taking same prior on hypothesis: $P(H_2|I) = P(H_3|I)$

Bayes factor

A third class of GRB?

$$O_{23} = \frac{P(D|H_2 I)}{P(D|H_3 I)}$$

To compute the Bayes factor we need the two evidences, the normalizations constants from before

We want to sample all the parameters space, not only the peak

Use Nested Sampling [NS]

$$P(D|H_i I) = Z_i = \int_{\Theta} P(\theta|I) P(D|\theta H_i I)\, d\theta = \int_0^1 \mathcal{L}(X)\, dX$$

From a 5D integral to a 1D integral between 0 and 1

Where $dX = P(\theta|I)\, d\theta$ so X is the prior mass

Likelihood as function of the prior mass

A third class of GRB?

We can chose the order in which accumulate prior mass, order by likelihood

$$X(\lambda) = \int_{\mathcal{L}(\theta)>\lambda} P(\theta|I)\, d\theta$$

We can think $\mathcal{L}(X)$ as the inverse function

$$\mathcal{L}\big(X(\lambda)\big) \equiv \lambda$$

In this way $\mathcal{L}(X)$ is a decreasing monotonic function with $\mathcal{L}(1) = 0$

A third class of GRB?

How does the algorithm work?

- Initialise:
  - N live points in parameters space
  - $\tilde{X}_0 = 1$
  - $Z = \sum_i Z_i$

$$\theta_1, \dots, \theta_N$$

- At each step i:
  - Evaluate $\mathcal{L}$ over this points
  - Sort the values in decreasing order and find the minimum, that will have greatest prior mass
  - Find width of the bin
  - Add to the evidence
  - Extract a new point with $\mathcal{L} > \tilde{\mathcal{L}}$

$$\mathcal{L}_1, \dots, \mathcal{L}_N$$

$$\min \mathcal{L} \equiv \tilde{\mathcal{L}} \Rightarrow \tilde{\theta},\ \tilde{X} \equiv X(\tilde{\theta})$$

$$w_i = \tilde{X}_i - \tilde{X}_{i-1}$$

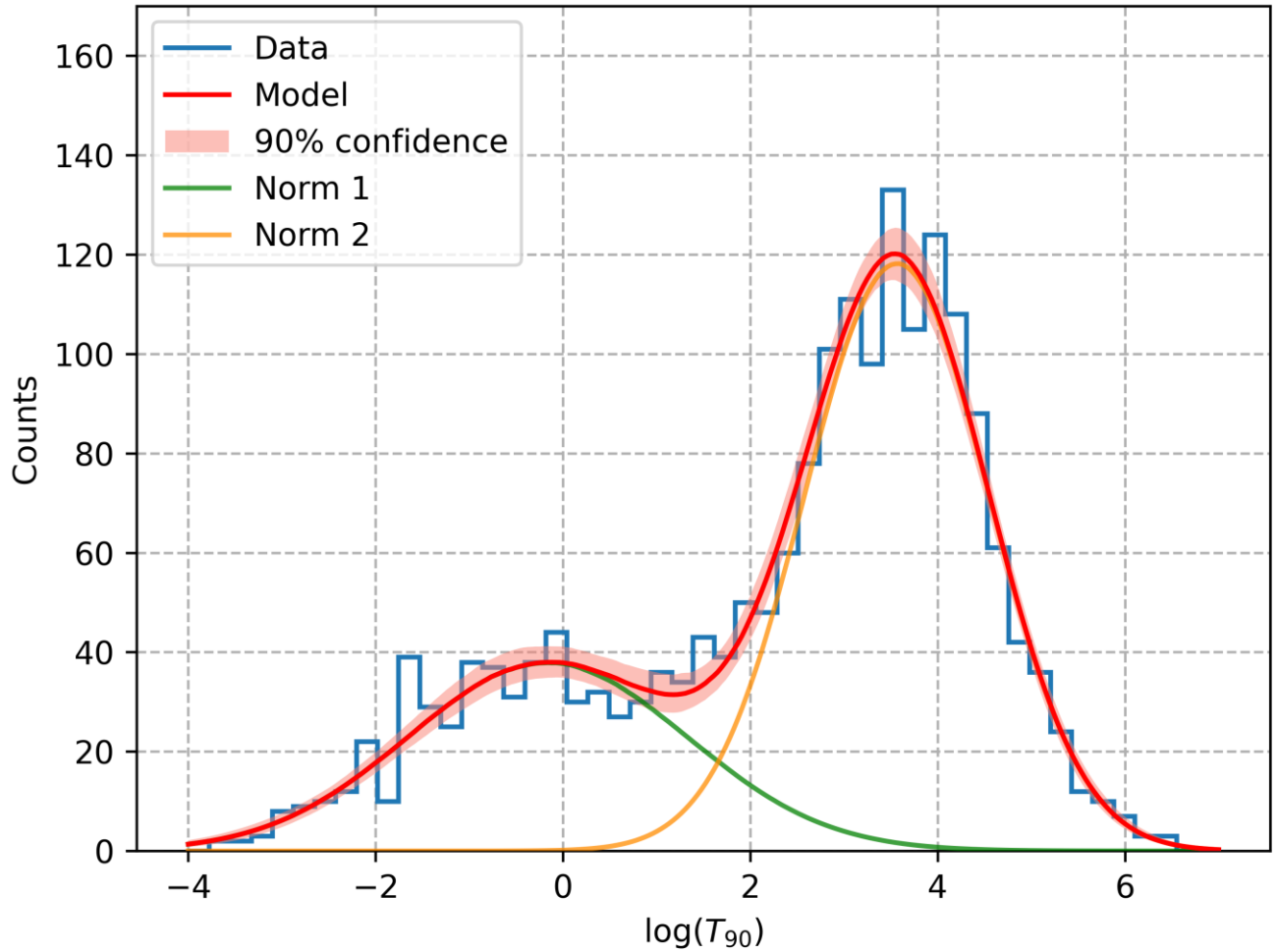
$$Z_i = w_i \tilde{\mathcal{L}}$$

By MCMC sampling

A third class of GRB?

Run algorithm with 5k live points with prior and likelihood as in the first inference

2 classes

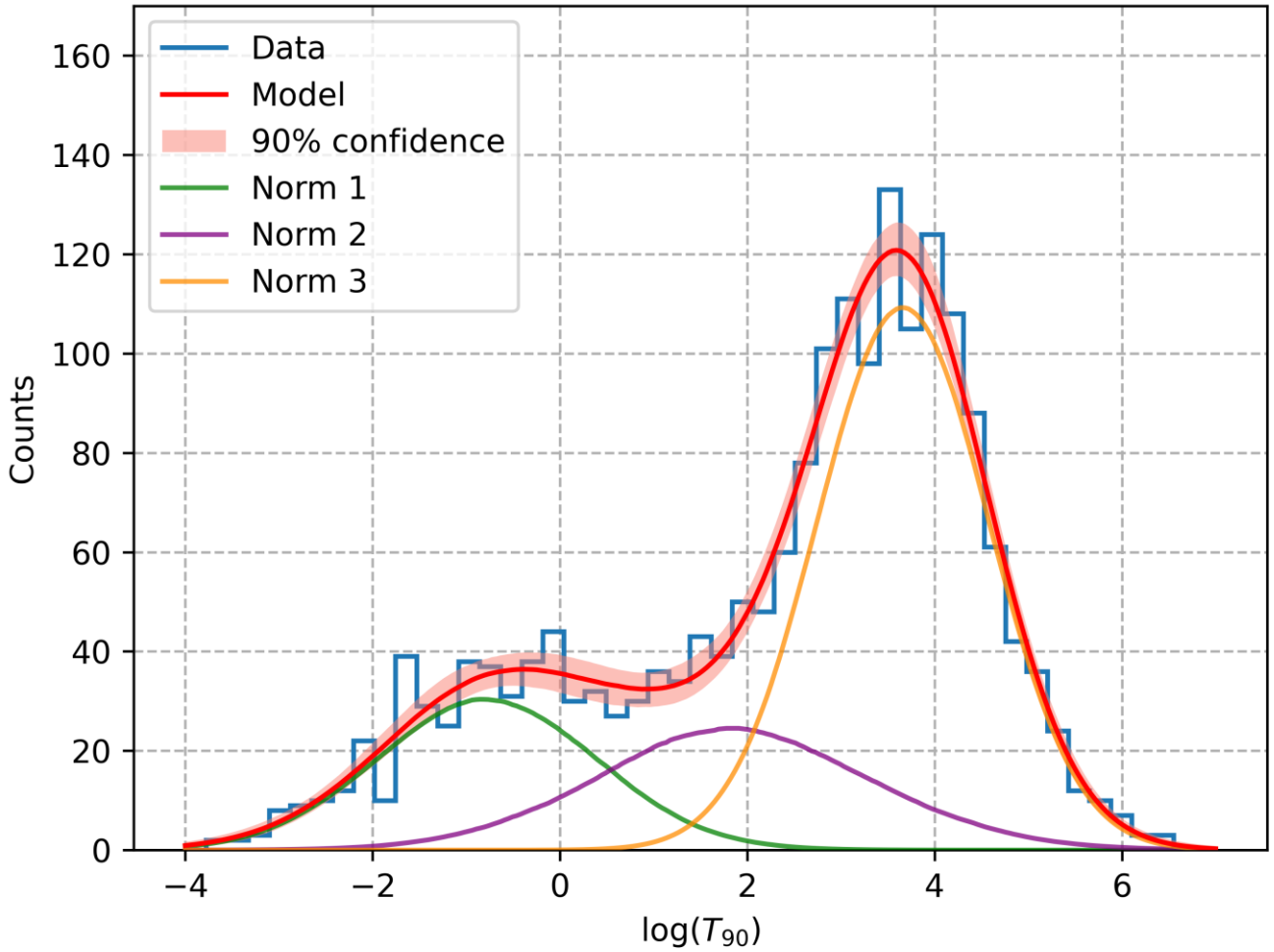


$\log(Z_2) \simeq 5899.82$

A third class of GRB?

Run algorithm with 5k live points with prior and likelihood as in the first inference

3 classes



$\log(Z_3) \simeq 5900.68$

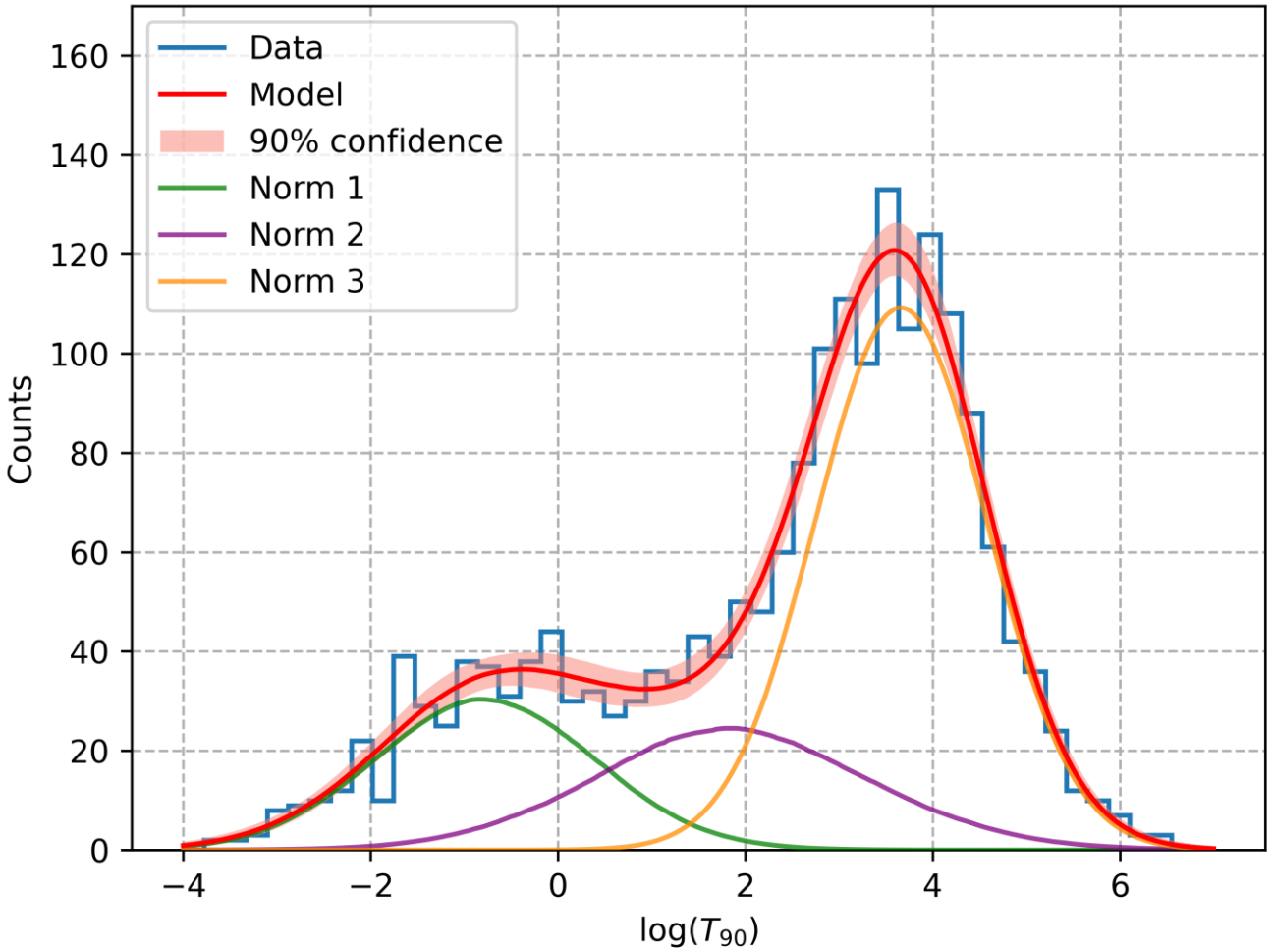
A third class of GRB?

$\log(Z_2) \simeq 5899.82$

$\log(Z_3) \simeq 5900.68$

$$O_{23} = \frac{Z_2}{Z_3} = 0.42$$

From the model and our dataset it's 40% more probable that there are 3 classes of GRB!



Conclusions

-

Bibliography??

Grazie per l’attenzione